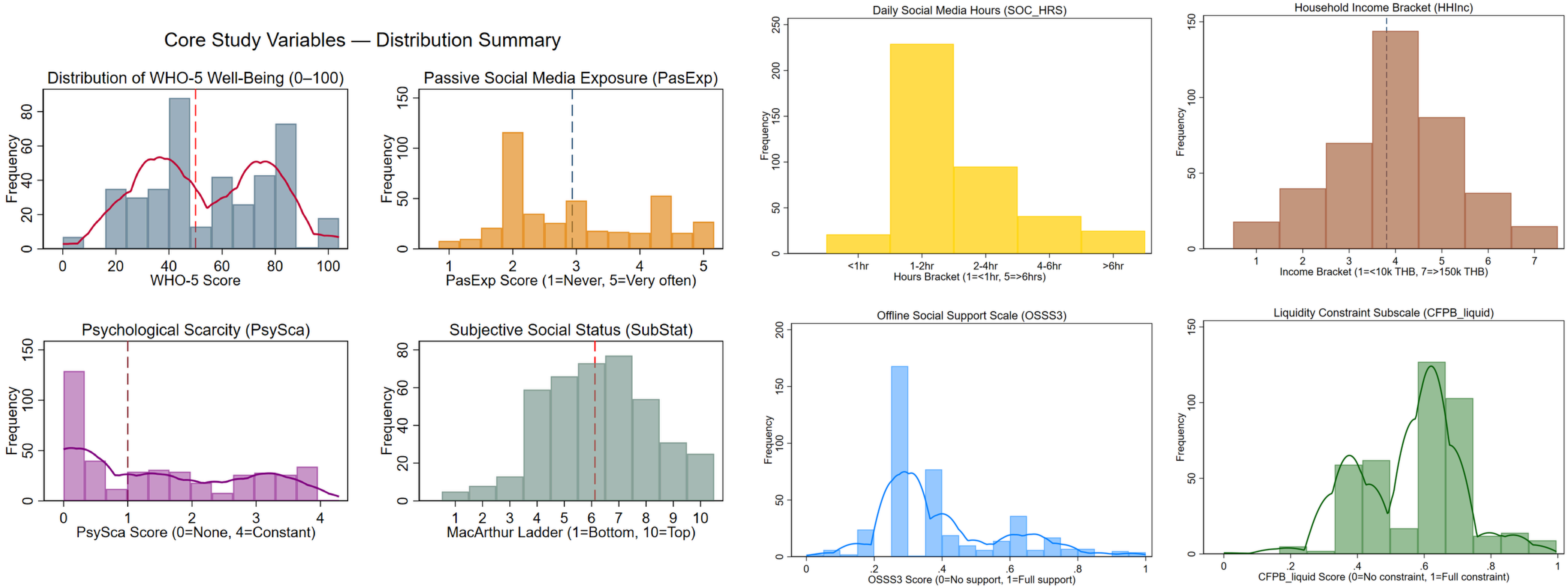
Behavioural Economics of Well-Being: Scarcity, Social Rank, and Media Exposure among Thai Adults in Bangkok

Main Idea

How they shape Thai urban's well-being?

Objective

- Does passive exposure moderates between subjective social staus and well-being?
- Does digital wealth intensify distress to those low perceived status group?



411 Observations in total

Structural Equation Model

$$\mathrm{WHO5}_i = \alpha + \beta_1(PsySca) + \beta_2(SubStat) + \beta_3(PasExp) + \beta_4(SubStat\ x\ PasExp) + \gamma X_i + \epsilon_i$$

M3 (dropped control lists and included interaction term)

$$\mathrm{WHO5} = 18.98 - 4.10 \cdot \mathrm{PsySca} + 10.21 \cdot \mathrm{SubStat} + 8.63 \cdot \mathrm{PasExp} - 2.68 \cdot (\mathrm{PasExp} \times \mathrm{SubStat})$$

Note: In M3, SubStat and PasExp denote conditional effects at fixed values of the interacted variable.

Marginal Effect of PasExp on WHO5

$$\partial \mathrm{WHO5} / \partial \mathrm{PasExp} = 8.627 - 2.678 \cdot \mathrm{SubStat}$$

- sample mean SubStat is 6.24,

Marginal Effect of SubStat on WHO5

$$\partial \mathrm{WHO5} / \partial \mathrm{SubStat} = 10.212 - 2.678 \cdot \mathrm{PasExp}$$

- sample mean PasExp is 2.942,

Base Model (M1) excluding WrkIndy_grp

heterskedasticity confirmed and justified (vce(robust))

```
. regress WHO5 ///
>     PasExp PsySca CFPB_liquid SubStat HHInc SOC_HRS ///
>     d_female d_gnd_nd d_age25_34 d_age35_44 ///
>     d_live_alone d_live_shared d_live_other, ///
>     vce(robust)
```

Linear regression

Number of obs	=	411
F(13, 397)	=	29.48
Prob > F	=	0.0000
R-squared	=	0.3692
Root MSE	=	19.225

WHO5	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
PasExp	-9.622453	.8873575	-10.84	0.000	-11.36696	-7.877946
PsySca	-3.677609	1.091766	-3.37	0.001	-5.823975	-1.531243
CFPB_liquid	-9.272908	8.945314	-1.04	0.301	-26.85902	8.313199
SubStat	3.267703	.9251906	3.53	0.000	1.448818	5.086588
HHInc	-.2127525	1.040292	-0.20	0.838	-2.257922	1.832417
SOC_HRS	2.240216	1.309489	1.71	0.088	-.3341834	4.814614
d_female	-1.26045	1.932484	-0.65	0.515	-5.059632	2.538732
d_gnd_nd	10.58192	5.31779	1.99	0.047	.1273701	21.03647
d_age25_34	3.270438	2.371898	1.38	0.169	-1.392613	7.933489
d_age35_44	.1712509	2.638397	0.06	0.948	-5.015725	5.358227
d_live_alone	.7290949	2.789257	0.26	0.794	-4.754466	6.212656
d_live_shared	-2.974133	2.853961	-1.04	0.298	-8.584899	2.636634
d_live_other	-1.030566	3.016187	-0.34	0.733	-6.960261	4.899129
_cons	66.58351	9.882917	6.74	0.000	47.15412	86.01291

- R^2 = 0.3692; F(13,397) = 29.48
- All independent variables are significant at ***$p < 0.001$.
- All control lists are not significant.

Base Model (M2) including WrkIndy_grp

```
. regress WHO5 ///
>     PasExp PsySca CFPB_liquid ///
>     SubStat ///
>     HHInc SOC_HRS ///
>     d_female d_gnd_nd ///
>     d_age25_34 d_age35_44 ///
>     d_live_alone d_live_shared d_live_other ///
>     ib1.WrkIndy_grp, vce(robust)
```

Linear regression

Number of obs	=	411
F(17, 393)	=	23.60
Prob > F	=	0.0000
R-squared	=	0.3739
Root MSE	=	19.252

WHO5	Coefficient	Robust std. err.	t	P>|t|	[95% conf.	interval]
PasExp	-9.475573	.8996702	-10.53	0.000	-11.24434	-7.706805
PsySca	-3.419744	1.096965	-3.12	0.002	-5.576398	-1.26309
CFPB_liquid	-8.918854	9.011468	-0.99	0.323	-26.63557	8.79786
SubStat	3.344766	.9374598	3.57	0.000	1.501703	5.187829
HHInc	-.2229832	1.08706	-0.21	0.838	-2.360164	1.914197
SOC_HRS	1.976724	1.314892	1.50	0.134	-.6083791	4.561826
d_female	-1.375268	1.956422	-0.70	0.483	-5.22163	2.471094
d_gnd_nd	9.621618	5.442518	1.77	0.078	-1.078474	20.32171
d_age25_34	2.873513	2.412801	1.19	0.234	-1.870098	7.617125
d_age35_44	-.1440904	2.672431	-0.05	0.957	-5.39814	5.109959
d_live_alone	.1263356	2.813923	0.04	0.964	-5.40589	5.658562
d_live_shared	-3.042795	2.889006	-1.05	0.293	-8.722634	2.637044
d_live_other	-1.036771	3.007944	-0.34	0.731	-6.950445	4.876903
WrkIndy_grp						
Public/Essential	.3590892	3.037929	0.12	0.906	-5.613536	6.331715
Commerce/Hosp	-1.170068	2.846464	-0.41	0.681	-6.766269	4.426133
Labor/Ag/Informal	-4.609625	3.401413	-1.36	0.176	-11.29687	2.077615
Unclassified	4.719225	4.641989	1.02	0.310	-4.407012	13.84546
_cons	66.90589	10.35534	6.46	0.000	46.5471	87.26468

- R^2 = 0.3732; F(13,393) = 23.60
- All independent variables are significant at ***$p < 0.001$.
- Addition of WrkIndy_grp does not significantly have associations to WHO5; all control lists remain insignificant.

Final Model (M3) including Interaction Term (PasExp x SubStat) and excluding all control lists

Uncentre mean

```
. regress WHO5 c.PasExp##c.SubStat PsySca, vce(robust)

Linear regression                               Number of obs     =        411
                                                F(4, 406)         =     116.99
                                                Prob > F          =     0.0000
                                                R-squared         =     0.4049
                                                Root MSE          =     18.465

------------------------------------------------------------------------------
                 |               Robust
            WHO5 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
-----------------+------------------------------------------------------------
          PasExp |   8.627488   3.286474     2.63   0.009     2.166859    15.08812
         SubStat |   10.21161   1.364014     7.49   0.000     7.530193    12.89302
                 |
c.PasExp#c.SubStat |  -2.678339   .5015185    -5.34   0.000    -3.664236   -1.692442
                 |
          PsySca |  -4.097367   .9921629    -4.13   0.000    -6.047785   -2.146949
           _cons |   18.98411    9.45514     2.01   0.045     .3969649    37.57125
------------------------------------------------------------------------------
```

- R^2 = 0.40; F(4,406) = 116.99
- SubStat, Interaction term, PsySca are significant at ****p*<0.001; PasExp without compute at sample mean is marginally significant at ***p*<0.01.

Centre mean

```
COEFFICIENTS:

. di "  a (constant):       " %7.3f b_CONST
  a (constant):       18.984

. di "  b1 (PasExp direct):   " %7.3f b_PAS
  b1 (PasExp direct):    8.627

. di "  b2 (SubStat direct):" %7.3f b_LAD
  b2 (SubStat direct): 10.212
```

```
. di "  b3 (PsySca):    " %7.3f b_CW
  b3 (PsySca):     -4.097

. di "  b4 (PasExp x SubStat): " %7.4f b_INT
  b4 (PasExp x SubStat): -2.6783
```

Take partial derivatives to compute at marginal effects

```
. regress WHO5 c.PasExp_c##c.SubStat_c PsySca, vce(robust)

Linear regression                               Number of obs     =        411
                                                F(4, 406)         =     116.99
                                                Prob > F          =     0.0000
                                                R-squared         =     0.4049
                                                Root MSE          =     18.465

------------------------------------------------------------------------------
                     |               Robust
                WHO5 | Coefficient  std. err.      t    P>|t|     [95% conf. interval]
---------------------+--------------------------------------------------------
            PasExp_c |  -8.081177   .7897732   -10.23   0.000    -9.633733   -6.528622
           SubStat_c |   2.332987   .7349539     3.17   0.002     .8881969    3.777777
                     |
c.PasExp_c#c.SubStat_c |  -2.678339   .5015185    -5.34   0.000    -3.664236   -1.692442
                     |
              PsySca |  -4.097367   .9921629    -4.13   0.000    -6.047785   -2.146949
               _cons |   58.91698   1.468423    40.12   0.000     56.03032    61.80364
------------------------------------------------------------------------------
```

- R^2 = 0.40; F(4,406) = 116.99
- PasExp_c at sample mean becomes significant at ****p*<0.001 with justified coefficient signs (β = -8.08).
- SubStat_c at sample mean remains significant at ****p*<0.001 with justified coefficient marginal effects (β = +2.33).
- Interaction term, Psysca remains significantly the same.

Note: CFPB_liquid was dropped due to polarity problem on CFPB subscales (cfpb_con negative-sign factor loading)

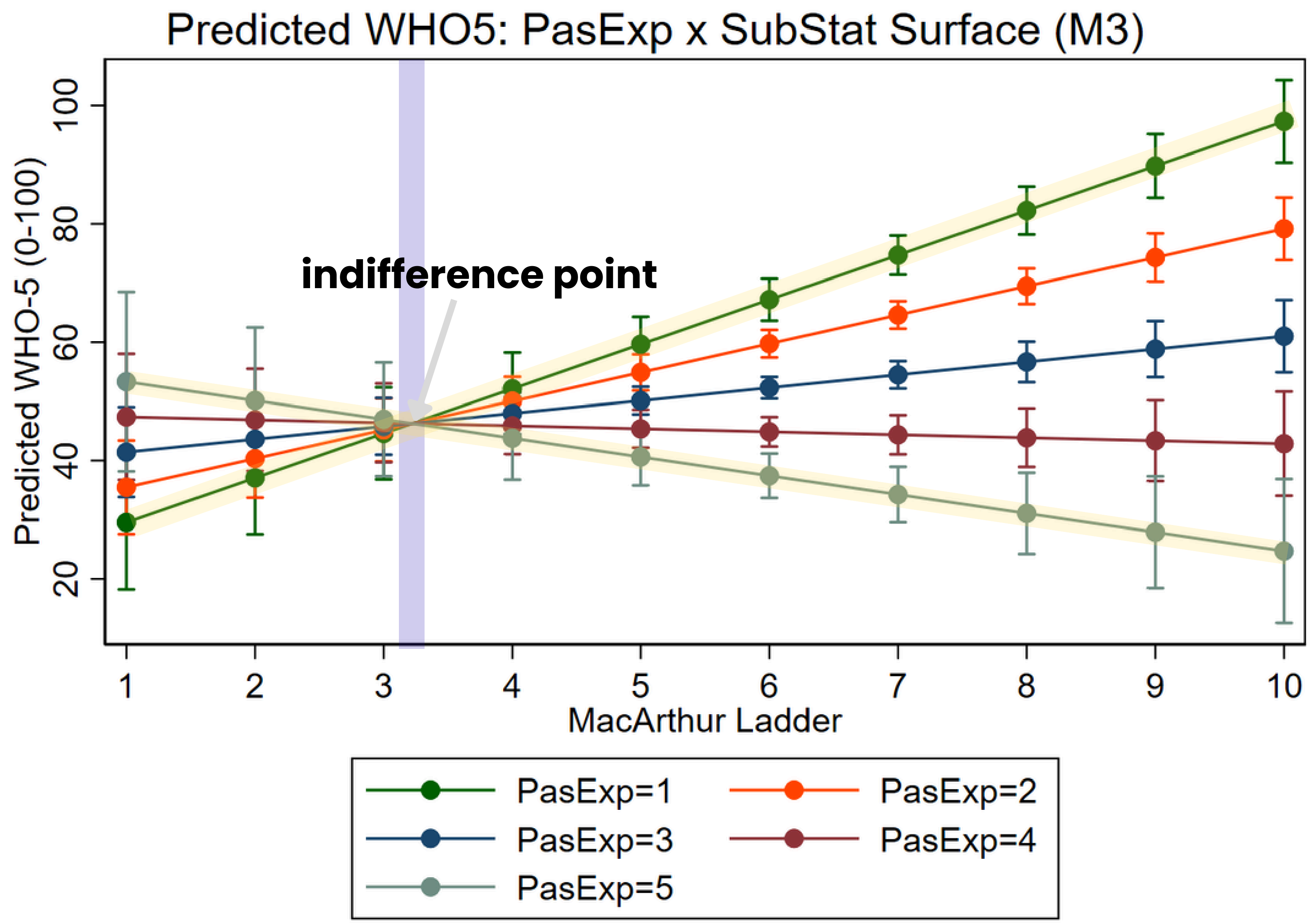
```
. estimates stats M1 M1r M3

Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df         AIC        BIC
-------------+---------------------------------------------------------------
          M1 |        411  -1885.762  -1791.074      14    3610.147   3666.408
         M1r |        411  -1885.762  -1789.553      18    3615.107   3687.442
          M3 |        411  -1885.762  -1779.089       5    3568.178   3588.271
-----------------------------------------------------------------------------
Note: BIC uses N = number of observations. See [R] IC note.
```

Evidence 1: Psychological Shield is crucial for protecting from passive exposure.

At higher SubStat levels, passive exposure moderates the individuals to either best possible mental well-being or worst possible mental well-being.



Note: assumes linear moderation; true zone may be non-linear and higher PsySca in low-status groups compounds their well-being disadvantage.

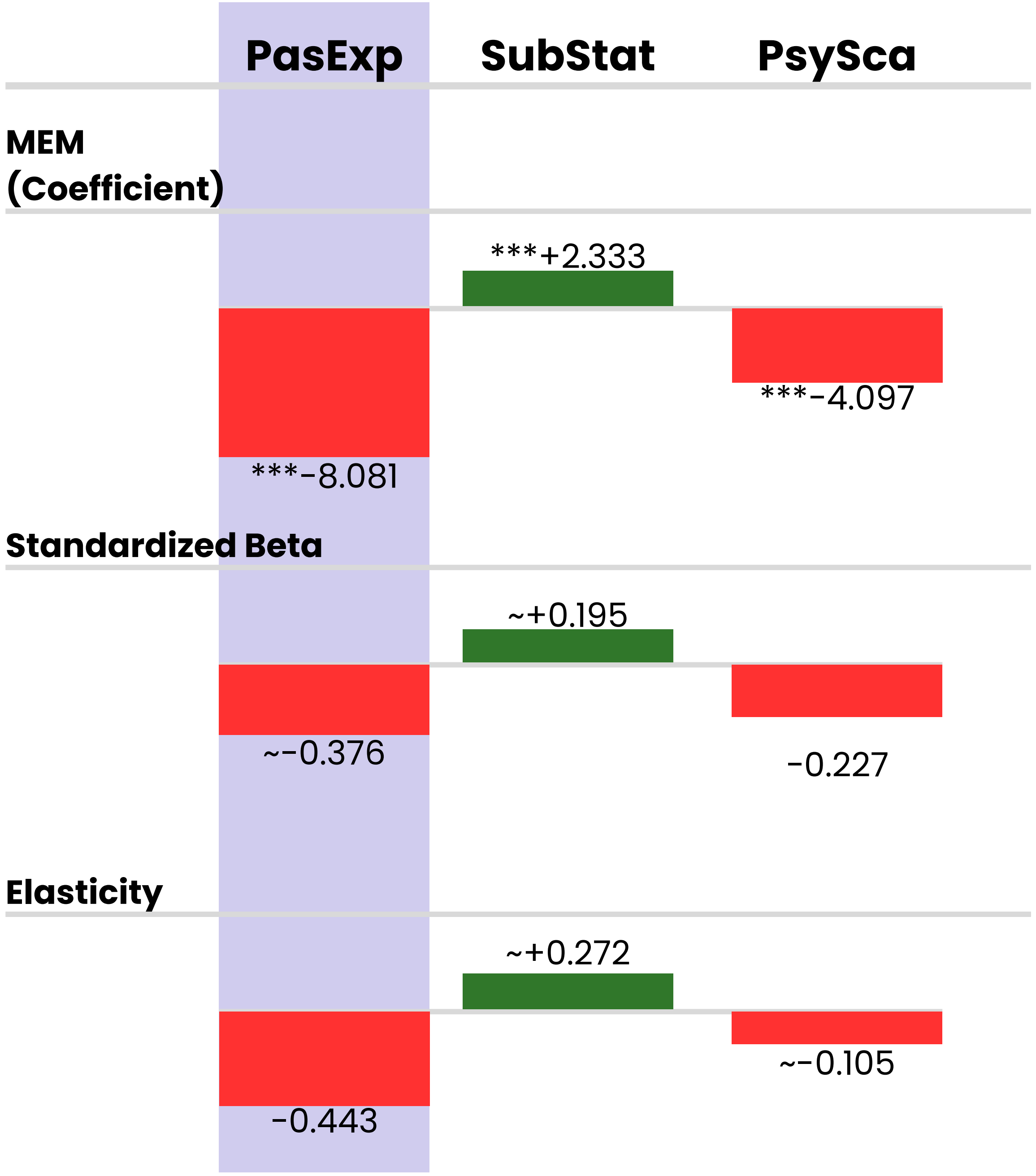
Key findings from Interaction Surface

- **Interaction coefficient**: ~-2.68 (95% CI: -3.664 to -1.692)
- **F-statistic**: $F(1, 406) = 28.52$
- **Indifference point**: Final derivative zero-crossing at SubStat ≈ 3.22

- **Prospect Theory:**
  - low status acts as digital vulnerability (Kahneman & Tversky, 1979; Skogen et al., 2022)
  - low status upward comparison ↑ → self-evaluation ↑ (Samra et al., 2022)
- **Festinger assimilation vs contrast (Festinger, 1954) :**
  - below zone → aspirational upward comparison (assimilation effect).
  - above zone → harmful contrast effect.

Evidence 2: Passive exposure is the most impactful driver of reduced well-being.

Regression, Standardized Beta, and Elasticity Analysis reveal that one change passive upward social media exposure affects the largest effects to mental well-being.



***p<0.001

Key Interpretation

- **PasExp**:
  - strongest driver of reduced well-being
  - intensive social media ↑ → relative deprivation ↑ , moderated by social class (Liu and Zhao, 2025).
- **SubStat:**
  - strong protective factor for well-being and outperform household income as a predictor,
  - social rank explains more, when objective income controlled (Tan et al., 2020; Navarro-Carrillo et al, 2020)
- **PsySca**:
  - moderate driver of reduced well-being.
  - cognitive bandwidth tax: scarcity ↑ → mental capacity ↓ → well-being ↓ (Mullainathan and Shafir, 2013).

Note:

1. Elasticity and standardized betas are approximations.
2. PasExp scale omits platform type and content valence; key moderators (Kross et al., 2021; Zhang et al., 2023).

Implications: Income and Subjective Status

Income gradient of subjective status is stable, with Cai et al. (2025) showing mediation via psychological security and Sekulova et al. (2023) confirming that income relief improves well-being and raises perceived social rank.

Household income bracket (1=<10k THB, 7=>150k THB)	high_status Low Statu	High Stat	Total
1	16	2	18
	88.89	11.11	100.00
	10.60	0.77	4.38
2	24	16	40
	60.00	40.00	100.00
	15.89	6.15	9.73
3	33	37	70
	47.14	52.86	100.00
	21.85	14.23	17.03
4	72	72	144
	50.00	50.00	100.00
	47.68	27.69	35.04
5	0	87	87
	0.00	100.00	100.00
	0.00	33.46	21.17
6	4	33	37
	10.81	89.19	100.00
	2.65	12.69	9.00
7	2	13	15
	13.33	86.67	100.00
	1.32	5.00	3.65
Total	151	260	411
	36.74	63.26	100.00
	100.00	100.00	100.00

Low income group (<10k THB/month)

- overwhelmingly low status (88.89%)

Middle income group (10k-50k/month)

- transitional, balanced status

Upper income group (>50k/month)

- nearly all high status (>86.67-100%)

Key Tabulations by Status Group

Industry sector vs. status

- High Status:
  - 42 (71.9%) Corporate/Professional
- Low Status:
  - 14 (46.67%) Labor/Agriculture/Informal
  - 43 (40.57%) Public/Essential

Scarcity by status at mean

- High status → 0.54
- Low status → 2.80

Offline support by status at mean

- High status → 0.37
- Low status → 0.44

Evidence

Implication

Key Effective Actions by Status Groups

To improve Bangkok adults' mental well-being, policymakers and public organizations should identify the social comparison indifference point as the basis for deploying subjective welfare most effectively.

Aspirational seekers (low status)

We should …

- **Offline activities + peer networks:**
  - ✓ comparison-free support (Zhang et al., 2023)
  - ✓ community-based resilience (Barua & Tejativaddhana, 2019; Samutachak et al., 2025)
- **Financial literacy:**
  - ✓ reframe perceptions can reduce financial scarcity (Guan et al., 2022)

We should not …

- **digital detox/shaming**

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Loss-aversion individuals (high status)

We should ...

- **Corporate wellness / media breaks**:
  - steepest negatve effects (ME(PasEx) = -9.15 at SubStat 6.64)
- **Cognitive Behavioural Therapy**:
  - low status mediates to mental health ↓ (Kraft & Kraft, 2023).
- **Algorithm literacy**:
  - key protective action (Zsila & Reyes, 2023)

We should not ...

- **ignore them**

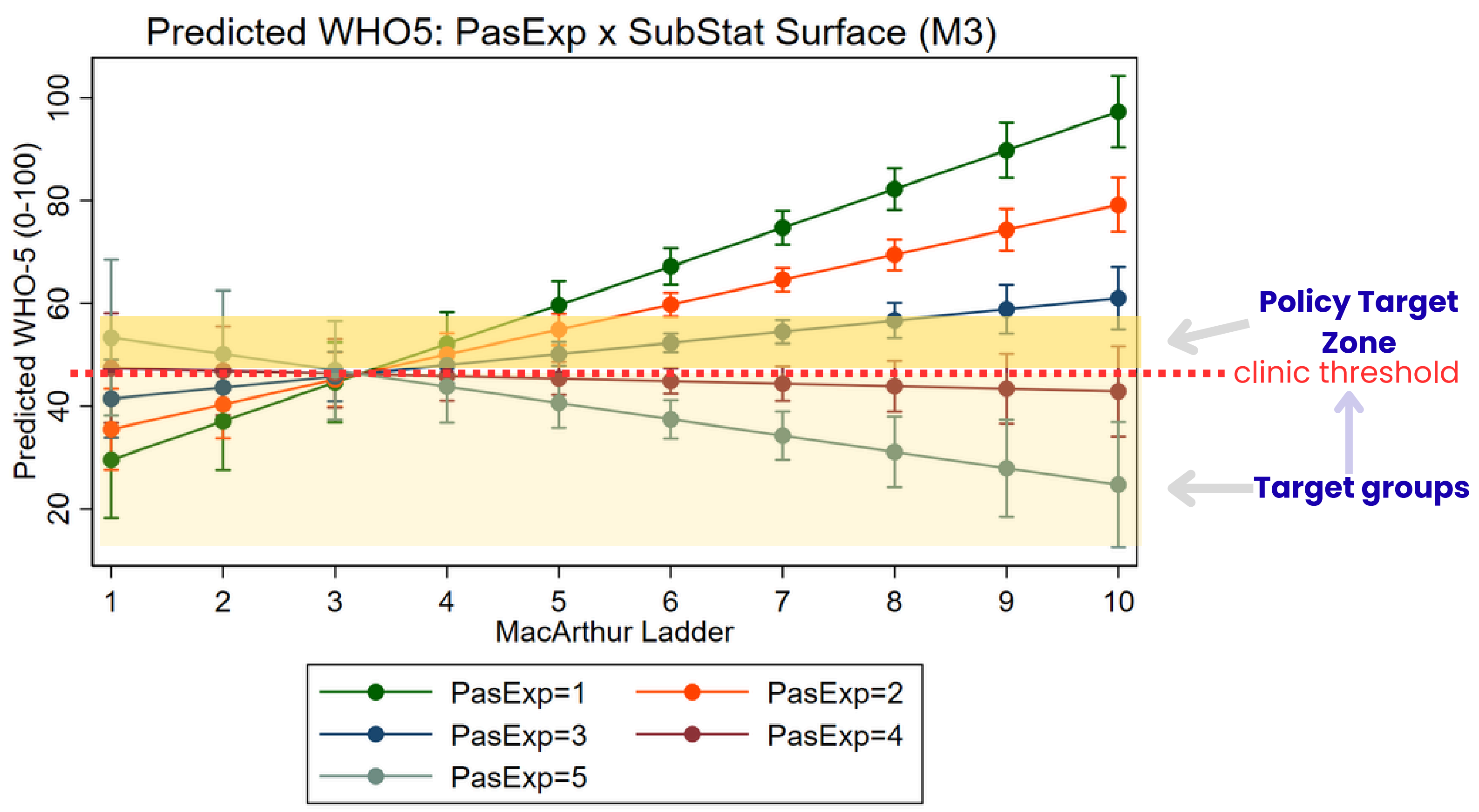
Policy Recommendation and Expected Impact

Implementations should target vulnerable groups below critical clinical threshold and emphasize the social comparison indifference point as the key leverage for effective welfare deployment.

Key Combinations:

- PasExp=1, SubStat=1 → predicted WHO-5 ≈ 29.5
- PasExp=5, SubStat=10 → predicted WHO-5 ≈ 13.4

Target vulnerable groups below WHO-5 threshold.



Actionable Recommendation

Expand income-based relief

- Expand urban relief (**welfare card**, **subsidies**, **debt restructuring**) based on household income to reduce scarcity and raise subjective status.

Deploying digital literacy program

- BMA should use local health centers to run digital well-being campaigns, **teaching residents about social media algorithms and management** to reduce passive exposure and improve mental health.

Integrate financial counseling in clinics

- Bangkok **clinics should partner with financial advisors to address depression and anxiety linked to financial stress and debt** among low-income groups.

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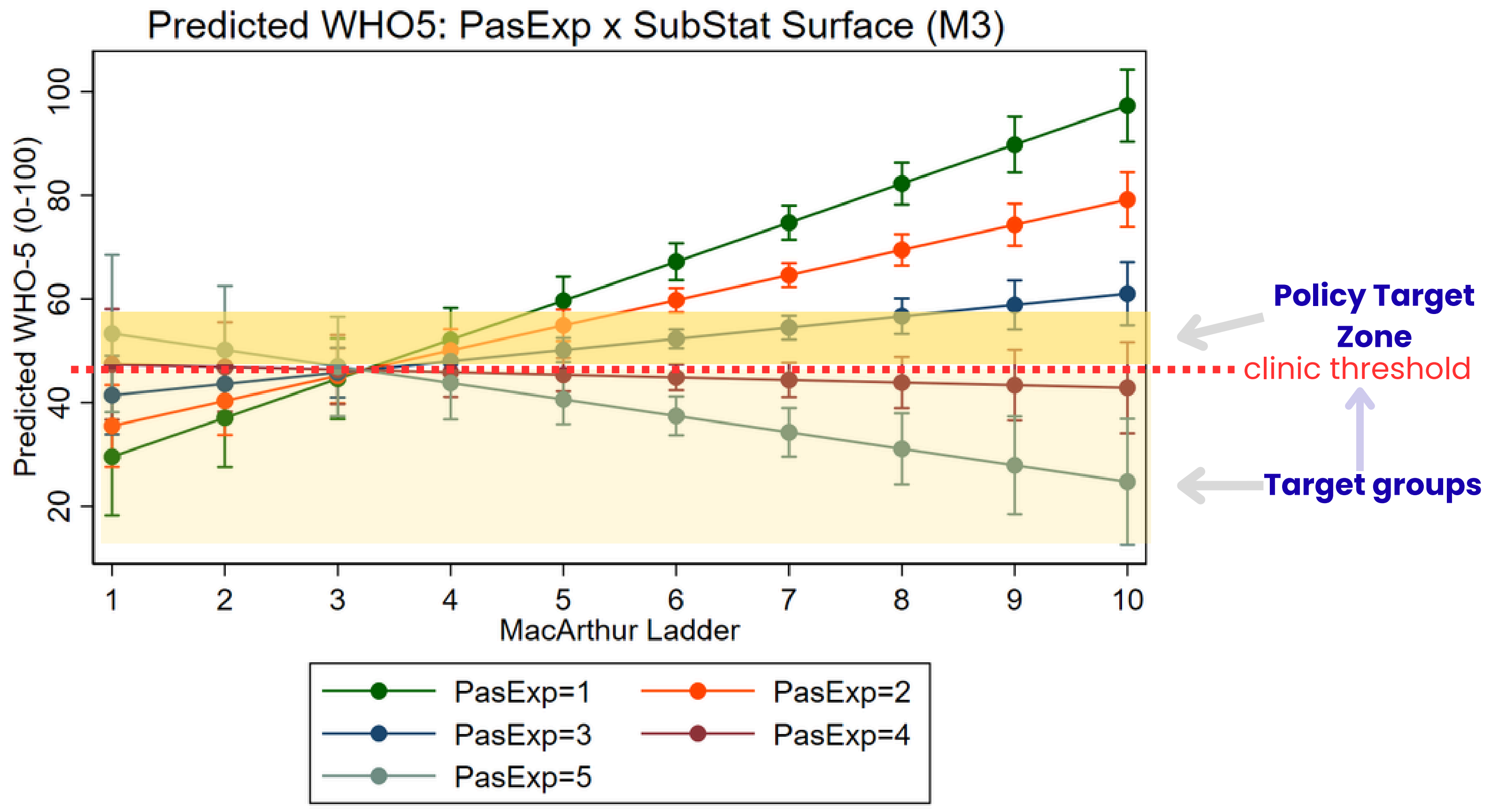

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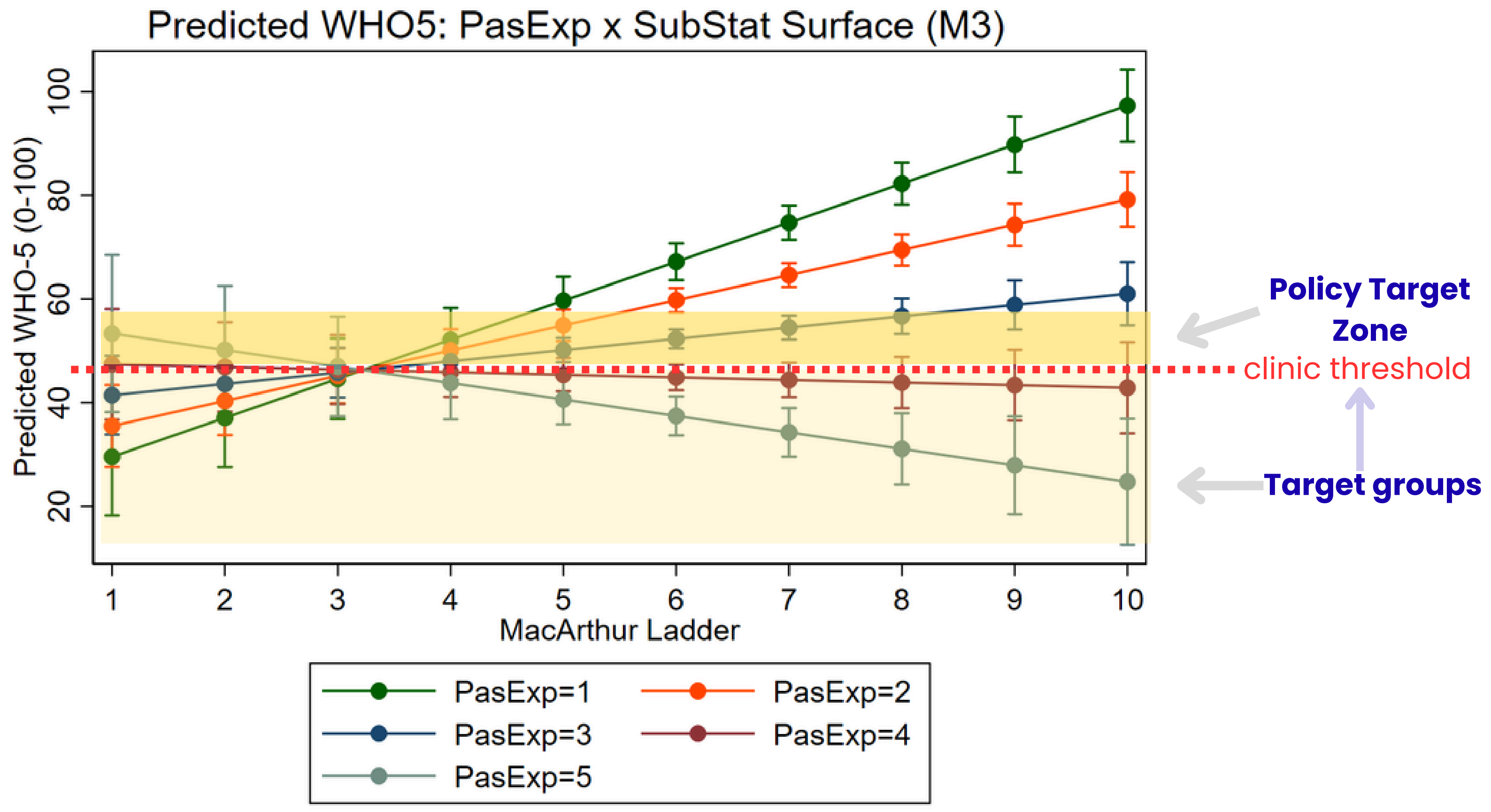

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Evidence

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Thank you

Group 3

Acknowledged Limitations

- cross-sectional study
- WHO5 bounded; cannot predict on extremes
- CFPB-subscale negative cfpb_con loaded
- overrepresents urban, digitally active adults
- platform choice and usage may be endogenous to well-being; Instrumental variable estimation recommended.

Robustness Check

- Heteroskedasticity confirmed (BP $p=0.031$, White $p<0.001$) → vce(robust) ✓
- Residual normality confirmed (SW $W=0.994$, $p=0.103$)
- VIF max 3.00 (Substat=3.00; mean VIF=1.63)
- Cook's D Sensitivity Check (22 influential obs, core results stable)
- Bootstrap 1,000 reps (BC-CI core independent variables not cross zero)